

Electoral Forecasting in Volatile Party System Settings: Assessing and Improving Pre-Election Poll Predictions in Italy

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Abstract

This study examines electoral forecasting in volatile party systems, focusing on factors contributing to deviations between poll predictions and actual election outcomes. Using Italy as a case study, it identifies biases in polling data and proposes a method to enhance estimator accuracy in a context of stable institutions and volatile electoral dynamics. Data from three Italian general elections are analyzed to evaluate discrepancies between pre-electoral polls and results, assessing key factors such as timing of data collection, survey methodology, sample size, and party system fragmentation. Employing a Bayesian inference process via a Markov chain Monte Carlo (MCMC) adaptive Metropolis-Hastings (MH) algorithm, the study demonstrates that pre-electoral estimates can be significantly improved using the Two-Stage Model (TSM). By consistently outperforming traditional poll predictions, the TSM offers a robust framework for addressing polling biases. These findings advance political forecasting by improving accuracy in both consolidated democracies and volatile electoral contexts, while emphasizing the need for future research on dynamic polling methods and fundamentals-based models.

Keywords

public opinion, pre-electoral polls, electoral forecasting, Italy, Bayesian inference

Introduction

In recent years, the reliability of pre-electoral polls has been increasingly questioned, especially in light of high-profile electoral mispredictions across the globe. Major electoral events, such as the Brexit referendum (2016) and the Australian federal election (2019), have demonstrated significant discrepancies between poll predictions and actual outcomes, raising doubts about the

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ability of traditional polling methods to accurately capture voter sentiment (Gelman, 2021; Jennings & Wlezien, 2018; Tao & Reis-Smart, 2014). The consequences of such inaccuracies are far-reaching. Not only do they undermine public trust in the democratic process, but they may also influence voter behavior, distorting the electoral landscape (Blais et al., 2009). If voters are consistently misinformed by flawed polling data, the risk of democratic processes being undermined increases (see Daoust, 2021), which calls for better electoral forecasting methods.

This paper explores the causes behind polling failures and proposes an innovative approach to enhance the accuracy of election forecasts. It focuses on Italy as a case study to address the challenges of forecasting in volatile party systems. Italy's political landscape is characterized by significant electoral volatility (Emanuele & Chiaramonte, 2020), a fragmented multiparty system (Pasquino, 2020), and shifting coalitions (Bartolini et al., 2004; Bergman, 2021), which makes it a challenging environment for pre-election predictions. The unusually high rate of electoral system turnover (Regalia, 2020) further complicates efforts to forecast election outcomes accurately. These dynamics provide a valuable context for examining how forecasting models might adapt and improve in such volatile settings. By studying the Italian context, this paper aims to bridge the gap between the forecasting methods used in more consolidated democracies, where political systems are stable, and emerging democracies, where party systems are more unpredictable.

The primary objective of this paper is to identify and understand the factors contributing to the discrepancies between pre-electoral polls and actual election results. By analyzing data from three Italian general elections (2008, 2013, and 2018), the study evaluates the accuracy of pre-electoral predictions and examines key factors that influence polling precision. These factors include sample size, the timing of data collection, survey methodology, and the level of fragmentation within the party system. The paper argues that while larger sample sizes are traditionally associated with improved accuracy, certain biases introduced by online polling methods can undermine this effect. Additionally, the paper investigates the impact of party system fragmentation on forecasting accuracy, hypothesizing that the greater the number of parties in competition, the more challenging it becomes to capture voter preferences accurately.

To address challenges, this study introduces a novel methodological approach: the Two-Stage Model (TSM). Initially developed for electoral forecasting in emerging democracies, the TSM applies a Bayesian inference process to improve forecast accuracy by adjusting for factors like polling reliability, sample size, and temporal proximity to the election (see Bunker, 2021). By incorporating these elements into a comprehensive modeling framework, the TSM provides a more robust tool for improving the accuracy of pre-electoral forecasts, particularly in environments marked by high levels of electoral volatility. This paper demonstrates the application of the TSM to Italy's electoral context and compares its performance to that of traditional polling methods.

The findings reveal several important insights. First, the paper confirms that polling accuracy improves when surveys are conducted closer to the election date, as voters are more likely to have made decisions in the final stages of the campaign. This finding aligns with prior research that shows the diminishing accuracy of forecasts made early in the election cycle. Second, the paper highlights the critical role of survey methodology in influencing polling outcomes. Mixed-mode surveys, which combine multiple data collection methods, consistently produce more accurate results compared to single-mode methods, such as online-only or phone-only surveys. Finally, the study demonstrates that the degree of fragmentation in the party system has a significant impact on forecasting accuracy. As the number of parties in competition increases, the complexity of capturing voter preferences rises, which in turn leads to greater discrepancies between poll predictions and actual results.

Most importantly, the TSM outperforms traditional polling methods, significantly reducing the mean absolute error (MAE) in forecasts and providing more accurate predictions for the majority

of competing parties. While the model was most effective in the 2013 election, where it improved upon the polls by an average margin of 1.4%, it also demonstrated a notable improvement in the 2018 election, even though the accuracy advantage was slightly less pronounced. These findings underscore the potential of the TSM to address the limitations of traditional polling methods, especially in multiparty systems characterized by electoral volatility.

The remainder of this paper is structured as follows. In Section 2, the theoretical framework for electoral forecasting is reviewed, focusing on the limitations of traditional polling methods and the role of emerging forecasting techniques. In Section 3, the Italian case is introduced, justifying its selection as an ideal context for studying the challenges of forecasting in a volatile party system. Section 4 outlines the hypotheses tested in the study, providing a clear rationale for the factors influencing polling accuracy. In Section 5, the data used in the analysis is described. In Section 6, the results of the accuracy assessment are presented, and in Section 7 the TSM is applied and compared to the performance of traditional polling methods. Section 8 follows with a discussion of the findings, and Section 9 concludes by highlighting the implications for future research in electoral forecasting.

Theory: Pre-Election Predictions

Voters rely on information to guide their electoral decisions (Cox, 1997; Markus & Converse, 1979). Political knowledge equips voters to navigate elections effectively, making them more likely to participate and choose candidates or parties that align with their preferences, thus maximizing their perceived utility (Bartels, 1996; Ghirardato & Katz, 2006). In contrast, uninformed voters are less likely to participate and, when they do, are less likely to select the “right candidate” (Pande, 2011), defined as the one they would have chosen if fully informed (Matsusaka, 1995). This knowledge deficit can distort electoral outcomes, reinforcing information biases and misinforming democratic processes (Blais et al., 2009). As a result, democracy itself can suffer, producing outcomes that fail to reflect the electorate’s true preferences (Fowler & Margolis, 2014).

Pre-electoral polls have long been the dominant method for tracking public opinion and predicting election outcomes (Banerjee, 1992; Graefe, 2014). Polls provide essential information on electoral trends, levels of competition, and the likely margins of victory (Skalaban, 1988). However, they are not merely descriptive tools; polls also influence voter behavior (Jennings & Wlezien, 2016). Voters exposed to polling data are more susceptible to phenomena such as the bandwagon or underdog effects, where knowledge of a candidate’s popularity or struggles may influence individual voting decisions (Morton et al., 2015). This dual role—informative and influential—has placed polls at the center of both public debates and academic scrutiny.

Yet, despite their prominence, polls are increasingly criticized for their failures to accurately predict electoral outcomes. High-profile polling errors, such as those during the Brexit referendum (2016) and Australia’s federal election (2019), have raised significant doubts about their reliability (Pennay et al., 2020; Sturgis et al., 2018). These failures have sparked investigations into the underlying causes of polling inaccuracies, which include sampling biases, late voter swings, and methodological limitations (Tao & Reis-Smart, 2014). As polling continues to face these challenges, there is growing interest in exploring alternative methods for predicting election outcomes.

Political science has responded to these concerns by devising innovative forecasting methods that address the limitations of traditional polls. Graefe et al. (2014) categorize these approaches into four main methods: polls, expert predictions, quantitative models, and betting markets. Each method has unique strengths and weaknesses. For instance, expert predictions, while historically valuable, have suffered from increasing susceptibility to wishful thinking and cognitive biases (Sjöberg, 2009). Similarly, betting markets, once viewed as a promising alternative, have faced scrutiny for structural biases and vulnerability to manipulation, which undermine their credibility

(Graefe, 2016). Emerging quantitative methods, such as those leveraging social media data and sentiment analysis, offer new possibilities but remain underdeveloped and prone to significant limitations (Juneja & Floridi, 2024; Jungherr et al., 2012; Tumasjan et al., 2010).

Among the proposed solutions, two forecasting approaches have emerged as particularly effective: fundamentals-based models and Dynamic Linear Models (DLMs). Fundamentals-based models rely on structural variables—such as economic conditions, incumbency, and historical voting patterns—to predict election outcomes. These models are regularly applied and particularly successful in consolidated democracies (Carlin et al., 2021; Love et al., 2024), where stable political systems and voter preferences allow long-term predictors to perform well (Graefe, 2016). The stability of institutions in these democracies ensures that key variables like inflation or unemployment have a consistent and measurable influence on voter behavior (Nadeau et al., 2020). As a result, fundamentals-based models have proven highly accurate in contexts where the political environment is relatively predictable.

In contrast, DLMs offer a more flexible and adaptable alternative. Grounded in Bayesian inference and state-space modeling, DLMs integrate polling data over time to estimate latent voter trends and project these trends into the future (West & Harrison, 1997). This continuous updating process allows DLMs to incorporate new information as it becomes available, making them particularly effective in contexts where voter preferences are dynamic or electoral environments are volatile (Walther, 2015). Their adaptability makes DLMs suitable for both consolidated and emerging democracies, where they have demonstrated success in a variety of electoral contexts.

In established democracies, DLMs have consistently proven their utility. Rigdon et al. (2009) applied Bayesian forecasting models to predict Electoral College outcomes in the United States, while Lock and Gelman (2010) used Bayesian inference to estimate vote shares for the 2008 U.S. presidential election. Linzer (2013) extended this by developing a dynamic Bayesian forecasting model that combined historical data with public opinion surveys to produce accurate vote-share estimates. In the United Kingdom, Fisher and Lewis-Beck (2015) used state-space models to forecast parliamentary seats, while Hanretty et al. (2016) employed similar methods to predict Westminster election outcomes and Hurtado Bodell (2016) and Stoltenberg (2013) further explored the application of DLMs in Nordic countries.

DLMs have also been successfully applied in newer democracies, where their flexibility enables them to account for high levels of volatility and uncertainty (Ratto et al., 2022). Bunker and Bauchowitz (2016) applied a DLM to multi-party electoral systems in Chile, highlighting their adaptability to diverse political contexts. Similarly, Cantu et al. (2016) integrated real-time polling data into Bayesian frameworks to improve forecast accuracy in Mexico. More recently, Bunker (2021) designed a DLM to predict election across 18 countries, demonstrating its ability to handle the unique challenges posed by fragmented party systems and volatile voter preferences, while Kang and Oh (2024) proposed a flexible Bayesian framework that shows promising results for the more established democracies in Asian. There are several recent examples of encouraging work in this line, in both established well studied countries (Chen et al., 2023; Hanretty, 2021) as well as new democracies in under studied regions (Kiingati, 2022).

But still, there remains significant room to move forward in addressing the “gray space” that exists between consolidated and emerging democracies. This less-explored area includes systems characterized by institutional stability but significant electoral volatility, such as frequent changes in party alliances or shifting voter coalitions. Neither fundamentals-based models nor DLMs have been systematically tested in these environments. Because fundamentals-based models have found some success in these sorts of democracies, DLMs, despite their adaptability, have not been rigorously applied. This presents a critical gap in the literature and an opportunity for future research to refine and extend these forecasting methods to better address the unique challenges posed by such intermediate electoral systems.

Case Study: Italy

This study focuses on Italy as a case study because it represents an interesting middle ground in the spectrum of democratic systems. While Italy is an established democracy, it has often been characterized by high levels of electoral volatility (Emanuele & Chiaramonte, 2020), making it a challenging context for forecasting. This volatility is particularly evident in its fragmented multiparty system (Pasquino, 2020; Regalia, 2020) and shifting coalitions (Bartolini et al., 2004; Bergman, 2021), which likely create difficulties in capturing voter preferences accurately. Studying polls in this environment offers valuable insights into party systems that fall between the extremes of stable, long-standing democracies and newer, unstable ones. By addressing this “gray space,” this study contributes to a deeper understanding of polling accuracy in complex and dynamic political systems.

It is also a great case to study, given that despite its status as an established democracy it relatively little has been done in the line of electoral forecasting. While there are several studies that look at the historic or specific accuracy of pre-electoral polls (Bellucci, 2010), it has mainly focused on evaluating specific elections (Hanretty, 2013). And while more recent literature has turned to more complex computational approaches including Bayesian methods, it has mainly focused on the impact of social media (Ceron et al., 2016a; 2016b), or on evaluating specific elements such as house effects (De Stefano et al., 2022). This gives us room to make a more general contribution to the electoral forecasting literature within the framework of a consolidated yet volatile democracies.

With such purpose, this study focuses on three specific election years: 2008, 2013, and 2018. They were chosen because they represent a full decade of Italian politics, marked by five-year intervals that break with previous and later elections. Together, they collectively form a political cycle that captures the ebb and flow of Italy’s electoral dynamics during a volatile period. While the 2008 election marked a turning point in Italy’s political history (Pasquino, 2009), elections after 2018 have been influenced by a series of simultaneous exogenous factors (Russo et al., 2023) that, at this point in time, seem more likely to introduce noise rather than signal into the research.

In this timeframe, 2008–2018, this study exclusively examines electoral competition patterns in the Chamber of Deputies, in which parties competed for 617 seats in each of the elections (not counting 13 reserved seats), albeit under different sets of electoral rules. While in the first two elections, votes were translated to seats using party-list proportional representation, in the third votes were translated to seats using a parallel voting system, in which 62.5% of the seats were allocated using party-list proportional representation and 37.5% of the seats were allocated using first-past-the-post (FPTP). In the case of 2018, this paper focuses only on the party-list proportional elections, since not all parties were on the national ballot for FPTP elections.

Hypotheses

Assessing the accuracy of public opinion polls requires examining the structural and methodological factors that influence polling outcomes. As widely established in the bodies of literature described above, polling accuracy is inherently tied to characteristics such as sample size, temporal distance from the election, and survey methodology, as prior research has shown. Larger sample sizes are generally expected to improve precision by reducing random sampling error, while polls conducted closer to election day tend to better reflect finalized voter preferences. Additionally, the method of data collection often introduces biases that can either enhance or detract from a poll’s representativeness.

This section builds on this conventional knowledge to propose four specific hypotheses regarding the drivers of polling accuracy in volatile systems. The first is based on the idea that larger

sample sizes reduce random sampling error, making the poll results more representative of the broader population. This is a well-established principle in survey research (see [Martin et al., 2005](#); [Samohyl, 2020](#)), where a greater number of respondents typically leads to more precise estimates, although there are reasonable doubts that this may still be the case ([Wang et al., 2015](#)). This gives us H1: as the sample size increases, polling accuracy improves (i.e., MAE decreases).

The second hypothesis is related to time (see [Jennings et al., 2020](#); [Sturgis et al., 2018](#)), and suggests that polls conducted closer to election day are more likely to capture finalized voter preferences, as many individuals only solidify their voting decisions in the final weeks leading up to the election, as polls become increasingly informative. Thus, polls fielded closer to election day, are expected to be more precise. From this we deduce H2: Polling accuracy improves as the temporal distance between the poll and election day decreases.

The third hypothesis stems from findings that closer inspect the relationship between methods and accuracy (see [Ansolabehere & Schaffner, 2014](#); [Dillman et al., 2014](#)), and suggests that mixed-mode surveys leverage multiple data collection methods, which improves the representativeness of the sample by reaching diverse demographic groups that may not be accessible through a single mode of collection. Following from above, H3 is: Mixed-mode polling methods yield more accurate results than single-mode methods (e.g., online-only or phone-only surveys).

Finally, the fourth hypothesis proposes that as the number of parties increases, so does the difficulty of accurately capturing voter preferences due to the fragmentation of voter support and the challenge of identifying smaller party supporters within a representative sample. Thus, we propose H4: The number of parties in competition negatively affects polling accuracy, as it increases the complexity of estimating voter preferences.

Data: A Decade of Italian General Elections

The main parameter of interest in this study is the difference between a party's performance in a general election and its corresponding pre-election poll prediction. To obtain results in terms of votes and seats for parties in an election, we referred to official data from the *Ministero dell'Interno*. For the latter, we put together a dataset of polling results that sought to capture electoral preferences derived from the survey question: "If the election were held today, for which of the following parties would you vote?" Here, we only included results from pollsters that conformed to the transparency and methodological rigor established by the *Dipartimento per l'Informazione nella Presidenza del Consiglio dei Ministri*. Only polls released prior to each election were included in the final dataset.

The analysis focuses on major parties, defined as those consistently securing at least three percent of voter preferences in pre-election polls. This threshold, while modest, is pivotal in parliamentary systems where smaller parties often influence post-election coalitions ([Golder, 2006](#)). Seats contested in reserved elections (e.g., regional or overseas) were excluded from the analysis. [Table 1](#) shows the ten parties that met this criterion. In 2008 and 2018, there were six major parties, while in 2013, the number increased to eight. Across all three elections, the major parties accounted for more than 90% of votes and 95% of the seats.

[Table 2](#), in turn, provides an overview of the pollsters included in the dataset and the characteristics of their methodologies across the three general elections analyzed in this study. A total of 27 pollsters met the inclusion criteria, which required adherence to national transparency standards and the public release of data prior to each election. The table reveals trends in the number and consistency of pollsters over time. While 16 pollsters fielded surveys in 2008 and 2018, only 14 did so in 2013. Importantly, only seven pollsters—Demopolis, Demos, Euromedia, IPR, Ipsos, Lorient, and Piepoli—conducted polls consistently across all three elections, while the

Table 1. Votes and Seats for Major Political Parties in Italian General Elections, 2008–2018

	2008		2013		2018	
	% Votes	#Seats	% Votes	#Seats	% Votes	#Seats
Sub-total major parties	91.93	617	92.18	617	94.38	617
Il Popolo della Libertà	37.38	272	21.56	97	14.00	103
Partito Democratico	33.18	211	25.43	292	18.76	107
Lega Nord	8.30	60	4.09	18	17.35	123
Unione di Centro	5.62	36	1.79	8	1.30	4
Italia dei Valori	4.37	28	2.25	0	—	
La Sinistra	3.08	0	3.20	37	—	
Movimento Cinque Stelle	—		25.56	108	32.68	226
Scelta Civica	—		8.30	37	—	
Liberi e Uguali	—		—		3.38	14
Fratelli d'Italia	—		1.96	9	4.35	32
Other parties	8.07	10	7.82	11	5.62	8
Vall d'Aosta (reserved)		1		1		1
Overseas (reserved)		12		12		12
TOTAL	100	630	100	630	100	630

Source: Authors with data from *Ministero dell'Interno*, see: elezionistorico.interno.gov.it/.
Note. Total corresponds to the sum of Sub-Total Major Parties and Others. Unione di Centro evolved into Noi con l'Italia; Il Popolo della Libertà later evolved into Forza Italia.

Table 2. Major Pre-election Pollsters in Italian General Elections, 2008–2018

	2008	2013	2018
Total pollsters	16	14	16
Total polls	53	168	147
Methods used	Phone, mixed-method	Phone, online, mixed	Phone, online, mixed
Average N	1119	1205	1323
Total N	57,290	179,470	178,107

Source: Authors with data from *Presidenza del Consiglio dei Ministri*, see: sondaggipoliticoelettorali.it/.
Note. The Table only considers pollsters fielded 365 days before election-night.

remaining pollsters participated in one or two elections. This variation underscores the dynamic nature of Italy’s polling landscape and the challenges of maintaining continuity in data sources. Pollsters employed three primary methods of data collection: phone, online, and mixed methods (a combination of phone, online, and occasionally face-to-face techniques). Mixed methods were more common in earlier elections, while online surveys grew increasingly prevalent in later ones. The average sample size (N) remained relatively consistent across elections, but the total sample size (total N) was notably higher in the two latter elections.

Accuracy Assessment: Testing the Conventional Wisdom

This section assesses the accuracy of pre-electoral polls in two steps. First, it evaluates the overall trends and patterns of polling accuracy across three elections, focusing on deviations from electoral results. Generally, these deviations have increased over time and are higher for individual

parties than for aggregated party groups. Second, it turns to the hypotheses to examine the specific factors contributing to these deviations by disaggregating the data into key variables such as sample size, temporal distance, and polling methods. This two-step analysis provides a comprehensive understanding of both the general trends and the underlying causes of polling errors.

There are different methods to measure pollster accuracy. Since they have been largely discussed in the literature, this paper will not go into detail. However, it is relevant to mention that most of the current measures track back to Mosteller's seminal work (1948). In their well-known report, they proposed eight measures to estimate poll accuracy. Crespi (1988) compared them using evidence from polls and elections in the United States and found no significant difference between them. Panagakis (1999) argues that for all the proposed measures, method number 3, or the mean absolute error (MAE), was marginally better to evaluate poll accuracy. As a result, the MAE has been used extensively in subsequent research (Bunker, 2021; Graefe et al., 2014; Jennings & Wlezien, 2016; Jowell et al., 1993; Kennedy et al., 2017; Traugott, 2005).

In the Italian context, it has also been used considerably. However, some of the most recent research has tended to use it interchangeably with the A-method, proposed by Martin et al. (2005). For example, Callegaro and Gasperoni (2008), Gasperoni (2013), and D'Agata and Tomaselli (2013) all prefer the latter. However, while the specification does indeed have advantages, none of their research shows significantly different results between the two measures. Consequently, this paper takes the simpler of the two approaches and uses MAE to estimate and compare accuracy.

Table 3 shows the MAE for polls predictions in the three elections. It differentiates between the MAE for all of the poll predictions made in the year leading up to the election (full sample) and the MAE for the last poll fielded by each of the pollsters (partial sample), is theoretically a better estimate, since it is closer to election-night. The results show that this intuition is correct. The MAE of partial sample tends to be more accurate than the MAE of the full sample.

Two other results are worth noting. First, that polls in the 2008 election were relatively more accurate than polls in the other two, with a MAE consistently below the three percent rule-of-thumb margin of error. The other interesting finding is that, with the exception of 2008, the prediction for the winner was worse than the overall prediction.

We now turn to address the four hypotheses outlined above. The dependent variable is Mean Absolute Error (MAE), and it is measured following the method explained above. The main independent variables are sample size (n), temporal distance (Δ), polling method, and the number of parties. Sample size was squared to account for diminishing returns, which is important particularly in the context of online polls. Early online polls introduced in 2013 often featured above-average sample sizes but were prone to inaccuracies due to representativeness issues (Hargittai & Karaoglu, 2018). Temporal distance is the number of days before an election,

Table 3. MAE (%) of Pre-election Polls in Italian General Elections, 2008–2018

	2008		2013		2018	
	All polls	Last polls	All polls	Last polls	All polls	Last polls
Parameter						
All candidates	2.4	2.1	4.2	3.0	3.1	2.8
Top-four	2.2	1.9	5.8	4.7	4.1	3.9
Winner	2.2	1.8	3.8	4.9	5.2	4.8
Runner-up	2.7	1.9	9.8	11.3	6.4	4.2
Third place	2.9	2.9	2.2	3.1	3.4	4.0

Source: Authors.

transformed to weeks and logged to address the disproportionate number of polls fielded early in the electoral cycle and Italy’s two-week pre-election polling embargo which limits the availability of polling data close to the election (Chung, 2012). Finally, polling method was recoded to reflect the made modes of data gathering and processing (Dillman et al., 2014). In the models, method can take one of three values: 0 = mixed, online and phone; 1 = online; 2 = phone. To test for hypothesis 4, an additional variable was added to account for party system fragmentation. This variable, party, is simply given by the number of parties in competition in a given year.

To test the hypotheses stated above, a fixed effects model was specified as follows

$$MAE_{it} = \beta_0 + \beta_1 n_{it} + \beta_2 time_{it} + \beta_3 method_{it} + \beta_4 parties_{it} + \gamma_i + \epsilon_{it}$$

In this equation, MAE represents the polling error for pollster i at time t . The predictors include the aforementioned squared sample size, logged temporal distance from the election, polling method, with the three categories mentioned above, and the number of parties. The term γ_i represents the fixed effect for pollster i , capturing time-invariant characteristics specific to each house, such as quality of execution, processing or geographic scope. Finally, ϵ_{it} represents the error term. By including γ_i , the models control for unobserved heterogeneity between pollsters, allowing the coefficients of interest (β_1 , β_2 , etc.) to reflect within-pollster effects. The models are fitted using maximum likelihood estimation with robust standard errors to account for potential heteroscedasticity. Model fit is evaluated using metrics such as Log Likelihood, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC).

Table 4 presents the results. The first three models focus on individual elections in 2008, 2013, and 2018, while the fourth combines data from all three elections. The fifth model builds on the combined data by including the number of parties as an additional predictor.

The results provide mixed support for the hypothesis that larger sample sizes improve polling accuracy (H1). While it is a well-established principle in survey research that larger sample sizes

Table 4. Determinants of the MAE in Italian General Elections, 2008–2018

Model	(1) 2008	(2) 2013	(3) 2018	(4) Combined	(5) M4 + Parties
Sample size (squared)	−0.000 (0.000)	−0.000 (0.000)	0.000*** (0.000)	0.000 (0.000)	0.000 (0.000)
Temporal distance (Log)	0.596*** (0.081)	0.942*** (0.050)	0.128*** (0.030)	0.616*** (0.044)	0.635*** (0.037)
Method: Online		0.690** (0.309)	−0.449* (0.251)	1.374*** (0.210)	0.345* (0.188)
Method: Phone	0.462 (0.332)	0.628*** (0.225)	−0.072 (0.110)	0.514*** (0.128)	0.072 (0.105)
Number of parties					0.593*** (0.045)
Constant	1.085** (0.368)	1.247*** (0.239)	2.810*** (0.117)	1.601*** (0.184)	−2.870*** (0.364)
Observations	51	169	147	367	367
Log likelihood	−42.325	−185.617	−65.604	−461.191	−392.763
Akaike Inf. Crit	96.65	385.234	145.208	936.381	801.527
Bayesian Inf. Crit	108.241	407.143	166.141	963.719	832.77

Dependent variable: MAE (%).

Note. The Table only considers pollsters that fielded polls 365 days before election-night. Key: $p < .01$: ***; $p < .05$: **, $p < .01$: ***.

reduce random sampling error, the fixed effects models do not consistently find a significant effect of sample size on MAE. In Model 4, which combines data from all three elections, sample size is significant ($\beta = -0.000$, $p < .05$), but the effect size is negligible, suggesting a very limited practical impact. Results in [Table A1](#) (in the [Appendix](#)), which includes interaction terms between sample size and polling methods, reveal an important nuance: for online polls, increasing sample size is associated with a decrease in accuracy, as indicated by the significant negative interaction term ($\beta = -0.00$, 001 , $p < .05$). This result suggests that while larger sample sizes generally improve accuracy, the representativeness issues inherent in online polling—such as underrepresentation of key demographic groups—become more pronounced as sample size increases, leading to reduced accuracy. In the 2013 model, sample size is significant but similarly small in magnitude, further reflecting these irregularities. In the other models, the effect of sample size is insignificant, indicating that once pollster-specific factors are controlled for, sample size alone does not explain much of the variation in MAE. These findings emphasize that while larger sample sizes may improve accuracy in certain contexts, their overall impact is muted, particularly for online methods, where interaction effects demonstrate diminishing returns or even negative effects on accuracy.

The second hypothesis, which posits that polling accuracy improves as the temporal distance to election day decreases (H2), is strongly supported by the results. Across all five models, the coefficient for time is highly significant ($p < .01$) and positive, indicating that as the temporal distance increases, polling accuracy decreases. This finding aligns with prior research showing that polls conducted closer to election day capture voter preferences more accurately, as many individuals solidify their voting decisions in the final weeks of the campaign. These results confirm previous findings in Italy ([D'Agata & Tomaselli, 2013](#)) and resonate with cross-national evidence presented by [Jennings et al. \(2020\)](#), who analyzed polling data from 338 elections across 44 countries and found that forecasts typically optimize accuracy two to three months before election day, depending on institutions arrangements. The coefficients for temporal distance ranges from 0.596 in the 2008 model to 0.942 in the 2013 model, highlighting the consistent and substantial role of time in shaping polling accuracy.

The hypothesis that mixed-mode polling methods yield more accurate results than single-mode methods (H3) is partially supported. Mixed-mode polls generally outperform online and phone polls across the models, but the magnitude and significance of these differences vary. In 2013, online polls are significantly less accurate than mixed-mode polls ($\beta = 0.690$, $p < .05$), reflecting the challenges of online sampling in its early stages. However, by 2018, the performance of online polls improves, with a negative but marginally significant coefficient ($\beta = -0.449$, $p < .1$), suggesting some progress in the accuracy of online methods. Phone polls show mixed results, performing worse than mixed-mode polls in 2013 ($\beta = 0.628$, $p < .01$) but exhibiting weaker and often insignificant effects in other years. These findings suggest that while mixed-mode surveys remain the most accurate method overall, polling method effects are influenced by broader trends in technology and data collection practices.

The final hypothesis, which suggests that the number of parties negatively affects polling accuracy by increasing the complexity of estimating voter preferences (H4), is strongly supported by the results of Model 5. The coefficient for the number of parties is positive and highly significant ($\beta = 0.593$, $p < .01$), indicating that for each additional party in the election, the MAE increases by approximately 0.59%. This finding highlights the difficulty of capturing voter preferences in fragmented party systems, where smaller and niche parties can be harder to identify within a representative sample. While this may sound obvious, it is a relevant finding in a context in which trust in polls increases parallel to the fragmentation of the system, potentially opening a new avenue to understand system related uncertainty.

To ensure the robustness of these findings, we conducted additional analyses to account for potential deviations from the baseline models and to test for additional effects. In terms of deviations, [Table A2](#) (in the [Appendix](#)) presents results from OLS models with robust standard errors (OLS-RSE) to account for potential heteroscedasticity. The results remain consistent with the fixed effects models, reaffirming the significance of temporal distance as a predictor and highlighting the nuanced effects of polling methods and the number of parties on polling accuracy. Importantly, the OLS results confirm that sample size exhibits a marginal impact on accuracy in the combined model and an even smaller effect in the election-specific models.

In the same vein, we introduced election fixed effects into the combined models, as shown in [Table A3](#) (in the [Appendix](#)). These models account for differences across election years, such as unique political contexts or varying levels of voter uncertainty. The inclusion of election fixed effects does not alter the key findings. However, these models do reveal that accuracy was systematically lower in 2013 and 2018 compared to 2008, as evidenced by the significant coefficients for the election year variables ($\beta = 0.794$, $p < .01$ for 2013 and $\beta = 0.632$, $p < .01$ for 2018). These results highlight the impact of contextual factors specific to each election but do not diminish the central role of pollster-specific fixed effects in explaining variations in MAE.

In conclusion, the fixed effects models provide robust insights into the determinants of polling accuracy. Temporal distance is the most consistent and significant predictor, reaffirming the importance of conducting polls closer to election day. Mixed-mode polls maintain an edge in accuracy, though online methods have shown improvement over time. The effect of sample size is limited and inconsistent, underscoring the importance of considering representativeness alongside sample size. Finally, the number of parties in competition emerges as a key driver of polling error, underscoring the challenges of forecasting in complex electoral systems. These findings highlight the nuanced interplay of methodological and system-level factors in shaping polling accuracy.

Accuracy Improvement: Fitting a Dynamic Linear Model

The previous section demonstrated how and why poll predictions deviated from electoral results. This section turns to the second question: can more precise estimators be computed? To answer this, we apply the two-stage model (TSM) proposed by [Bunker \(2021\)](#) to forecast the same set of elections as those analyzed earlier and compare the results. The TSM has been shown to achieve significant levels of accuracy in forecasting elections in new democracies ([Bunker, 2021, 2022](#)). Its flexibility allows for adaptation, making it well-suited to address the challenges of forecasting elections in a volatile, multiparty democracy like Italy. This adaptability positions the TSM as a useful tool in contexts characterized by fluctuating party competition and voter uncertainty.

The TSM uses two sequential stages to process polling data and generate a forecast. The purpose of the first stage is to process the input data and define weights for each poll. Essentially, it weighs them according to three criteria: their accuracy track record (analogous to the House Bias), random error (measured with N), and their distance from the election (Δ). In turn, the purpose of the second stage is to generate electoral estimates (in line with previous works such as [Jackman, 2005](#); [Linzer, 2013](#)). To do this, it sequentially takes the weighted polls and enters them through a Bayesian inference process, in which posterior likelihood distributions are simulated through a Markov chain Monte Carlo (MCMC) adaptive Metropolis-Hastings (MH) algorithm.

To provide more detail, the first stage of the TSM computes a pollster rating using a baseline election. For each pollster, only the result of their last poll is compared to the actual election result. Based on the accuracy of this prediction, each pollster is ranked, and these rankings are then normalized to generate a score ranging from 0 to 1. This normalized score reflects the historical reliability of each pollster.

For each subsequent election, three factors are used to determine a final weight for each poll: the pollster rating, the temporal distance from the election, and the sample size. The temporal distance is given a score between 0 and 1, with closer polls to the election date receiving higher scores. Similarly, the sample size is also scored between 0 and 1, with larger sample sizes being weighted more heavily. These three scores—pollster rating, temporal distance, and sample size—are then aggregated to create a unique weight for each poll.

The final weight for each poll is computed as the average of these three components

$$w_i = (r_i + t_i + s_i)/3$$

where w_i represents the final weight assigned to poll i , r_i is the pollster rating based on historical accuracy, t_i is the temporal distance score, and s_i is the sample size score. This weighted average ensures that each poll's influence on the forecast is determined by its historical performance, closeness to the election, and sample size.

The second stage processes the weighted polls using a Bayesian inference framework to estimate the posterior distribution of vote share V_j for each party j . The posterior is defined as

$$P(V_j|Z) \propto P(Z|V_j) \cdot P(V_j)$$

This equation combines the prior belief about vote shares $P(V_j)$ with the likelihood $P(Z|V_j)$, based on the polling data (Z). The posterior is estimated using a Markov Chain Monte Carlo (MCMC) algorithm. Specifically, an adaptive Metropolis-Hastings (MH) method is employed, which iteratively refines estimates of V_j by sampling from the posterior distribution. The MCMC process is expressed as

$$\theta^{(t+1)} = \theta^t + \epsilon_t$$

where $\epsilon_t \sim N(0, \sigma^2)$ is the proposal distribution, and $\theta^{(t)}$ represents the parameter value at iteration t . The adaptive nature of the algorithm ensures that the sampling process efficiently explores the parameter space, reducing computational overhead while improving convergence.

As part of the adaptation process, the 2008 general election serves as a calibration event, establishing baseline pollster ratings for accuracy and reliability. Once this baseline is defined, the 2008 data is discarded, and the TSM is applied to the polling data from 2013 and 2018 to produce forecasts. The primary objective, then, is to estimate the vote shares for each of the major parties and assess the TSM's ability to improve upon traditional polling predictions. If the TSM demonstrates greater accuracy, it provides compelling evidence that voters could have received more reliable electoral information during these cycles.

The results presented below were tested under a variety of contexts to ensure robustness and accuracy. The script allows researchers to replicate the analysis using their own data or to test the model under different conditions. In the application of the TSM, the burn-in period was set to 10,000 iterations to ensure convergence of the posterior distribution, with an additional 40,000 iterations used for sampling. The proposal variance and learning rate were dynamically adjusted within the adaptive Metropolis-Hastings framework to optimize the efficiency of the MCMC algorithm. The parameters can be modified to explore different assumptions or adjust the model for different types of elections.

Table 5 compares poll predictions and the TSM forecast for the 2013 and 2018 elections, considering the full sample (all polls fielded within one year of election night). The results indicate that in both elections, the TSM was more accurate than the polls. While the TSM achieved a MAE below three percentage points, the polls consistently produced a MAE above this threshold. Moreover, the TSM generated a more accurate forecast for 75% of the competing parties. When

Table 5. Full Sample: Poll Predictions Versus TSM Forecast.

Year	Party	Result	Poll MAE	TSM MAE	Difference	TSM more accurate?
2013	Movimento Cinque Stelle	25.56	3.78	4.86	−1.08	No
	Partito Democratico	25.43	11.25	9.72	1.54	Yes
	Il Popolo della Libertà	21.56	3.06	2.15	0.91	Yes
	Scelta Civica	8.30	5.00	0.66	4.34	Yes
	Others	7.82	4.75	0.90	3.85	Yes
	Lega Nord	4.09	1.69	0.85	0.84	Yes
	Sinistra	3.20	2.01	0.83	1.17	Yes
	Rivoluzione Civile	2.25	2.71	2.05	0.66	Yes
	Unione di Centro	1.79	3.35	1.72	1.63	Yes
	Average MAE	—	4.18	2.64	1.54	Yes
2018	Movimento Cinque Stelle	32.68	5.08	4.85	0.23	Yes
	Partito Democratico	18.76	6.42	5.46	0.96	Yes
	Lega Nord	17.35	3.41	3.80	−0.39	No
	Forza Italia	14.00	1.66	1.71	−0.05	No
	Others	9.48	2.24	1.65	0.60	Yes
	Fratelli d'Italia	4.35	0.54	0.44	0.10	Yes
	Liberi e Uguali	3.38	2.61	2.70	−0.09	No
	Average MAE	—	3.14	2.94	0.19	Yes

Source: Authors.

the TSM was more precise (for 12 out of 16 parties), it was more accurate by an average of 1.4%. Conversely, when the polls were more precise (for 4 out of 16 parties), they were more accurate by only 0.4%. In other words, the TSM outperformed the polls by an average margin of approximately one percentage point.

Focusing solely on the 2013 election, the TSM was more accurate than the polls in predicting the final result for 8 out of 9 parties. It was particularly effective in forecasting the vote share for *Scelta Civica* (the fourth-place party, which secured 8.3% of the vote and played a decisive role in government coalition formation), achieving a precision advantage of over 3%. The only exception was the newcomer party *Movimento Cinque Stelle*, where the TSM underperformed in predicting the vote share. Turning to the 2018 election, the TSM was more accurate than the polls in forecasting the results for 4 out of 7 parties. Although it was less efficient than in 2013, the TSM's margin of advantage remained significant, with its average forecast still remaining below the three percent margin.

While the TSM does indeed provide a more accurate forecast than the polls, the question arises as to whether its precision margin can be further improved. The previous section demonstrated that data from the partial sample (i.e., the last poll) tends to be more accurate than data from the full sample (i.e., one year). This section examines that idea further. [Table 6](#) compares poll predictions and the TSM forecast for the 2013 and 2018 elections when considering the partial sample (only the last poll fielded by each pollster). Remarkably, the results suggest that in both elections, the TSM forecast was more accurate than the poll predictions. In both cases, the TSM achieved a MAE under 3%. Notably, the TSM's advantage over the polls was even greater in this scenario, delivering more precise forecasts for 15 out of the 16 parties.

When examining the distribution of results by party, the findings are once again remarkable. In this case, the TSM outperforms the poll predictions by 0.61% in 2013 and by 0.16% in 2018. While this may appear to be a slim margin, it is significant, particularly given that the TSM is now competing against the polls' best possible attempt to predict the election. In other words, even if the polls were arranged and combined in any consistent manner, no better prediction could be

Table 6. Partial Sample: Poll Predictions Versus TSM Forecast.

Year	Party	Result	Poll MAE	TSM MAE	Difference	TSM more accurate?
2013	Movimento Cinque Stelle	25.56	4.60	4.49	0.12	Yes
	Partito Democratico	25.43	10.39	9.52	0.87	Yes
	Il Popolo della Libertà	21.56	2.04	1.72	0.32	Yes
	Scelta Civica	8.30	2.29	0.42	1.87	Yes
	Others	7.82	3.11	1.84	1.27	Yes
	Lega Nord	4.09	1.46	1.10	0.36	Yes
	Sinistra	3.20	1.11	0.88	0.23	Yes
	Rivoluzione Civile	2.25	1.78	1.62	0.15	Yes
	Unione di Centro	1.79	2.04	1.74	0.31	Yes
	Average MAE	—	3.20	2.59	0.61	Yes
2018	Movimento Cinque Stelle	32.68	4.70	4.88	−0.18	No
	Partito Democratico	18.76	4.30	4.22	0.08	Yes
	Lega Nord	17.35	3.90	3.85	0.05	Yes
	Forza Italia	14.00	2.50	2.32	0.19	Yes
	Others	9.48	1.49	0.64	0.85	Yes
	Fratelli d'Italia	4.35	0.53	0.45	0.08	Yes
	Liberi e Uguali	3.38	2.46	2.39	0.07	Yes
	Average MAE	—	2.84	2.68	0.16	Yes

Source: Authors.

achieved. Compared to the full sample, the TSM improves its accuracy for eight parties in the partial sample. The most intriguing result from Table 6 is that the only party the TSM fails to forecast in 2018 is the same party it was unable to forecast in 2013: *Movimento Cinque Stelle*.

Discussion

The previous two sections assessed the accuracy levels of pre-electoral polls in three Italian general elections and proposed a method to increase the precision of estimates using the same set of data available to voters. The main takeaway is that while pre-electoral polls have been increasingly inaccurate as time has passed, margins of precision can nonetheless be increased. This is a relevant finding in the context of the particular case study, which has shown mounting levels of electoral volatility over the years. Thus, in light of the overall trend of deviations between poll predictions and electoral results, the contribution of the TSM is relevant in its own right, as it paves the way for future research to further increase the precision of pre-electoral estimates. In this endeavor, the findings made in both the assessment as well as the improvement sections provide crucial insight.

The former showed that two variables were especially relevant in explaining the accuracy of an average poll: the number of weeks between a poll and an election, and the data collection method. These findings are consistent with previous research (D'Agata & Tomaselli, 2013). Interestingly, however, it also showed that sample size was not as relevant as expected. This is presumably due to the debut of online polling in the 2013 election, as analysis in the Appendix confirms.

Perhaps the most interesting finding, however, was the positive and significant influence of the number of parties on the MAE. This study finds that as the number of competing parties increases, so does the error. This seems to be particularly true when the number of smaller parties increases, as their lower overall vote shares often reflect nuanced and dispersed voter preferences. The idea is that even in the face of a constant sample size, capturing these smaller, fragmented voter segments becomes increasingly difficult, leading to greater errors in poll predictions. As such, the inclusion of a larger number of parties inherently adds complexity, which, in turn, contributes to a higher

MAE. This finding reinforces the notion that the challenges of polling are not merely structural but also context-dependent, especially in multiparty systems where voter fragmentation is prevalent. This is a remarkably relevant, especially given the understudied relationship between the two variables in forecasting literature.

In comparison, the findings regarding sample size are less expected but still not entirely surprising. In typical sampling research, larger sample sizes generally result in greater accuracy. However, the results here suggest otherwise. One possible explanation put forward is the rise of online polling, which debuted in Italy for the 2013 election. Online surveys, while efficient and scalable, may struggle to capture harder-to-reach demographics, thereby reducing accuracy despite larger samples. Another potential factor could be the diminishing returns of larger sample sizes in the specific context of electoral forecasting, where other factors such as temporal proximity to the election and methodological differences may play more decisive roles.

In turn, the improvement section showed that even in the context of declining poll accuracy levels and increasing electoral volatility indices, the precision of pre-electoral estimates can still be improved. After adapting a model tested elsewhere, under different institutional but similar electoral conditions, to forecast elections, this paper demonstrated that the same data could be used to further increase the precision of forecasts for the majority of parties in competition (in comparison to poll predictions). When the TSM was adopted and compared to the best possible estimation of the polls, the former was superior. Indeed, of the 16 parties the two estimators were matched on, the TSM produced a superior forecast for 15 of them. The only party consistently difficult to forecast was the *Movimento Cinque Stelle*, in which case the error is presumably correlated with the party rather than with the model.

Conclusion

The global failures of polls to accurately predict recent high-profile elections underscore critical questions about their biases and overall utility. This article has sought to address two of these questions through the lens of Italy, an established multiparty democracy with volatile electoral dynamics. First, we asked: why do polls deviate from electoral results? Our findings underscore that two factors are paramount in explaining these deviations: the temporal distance between a poll and the election, and the methodological approach taken by pollsters. Specifically, polls conducted closer to election day and those employing mixed-method data collection consistently achieve greater accuracy. Conversely, while larger sample sizes are typically expected to improve precision, our study highlights how this relationship is disrupted in the context of online polling due to biases in representativeness.

Second, we explored whether it is possible to generate more accurate electoral estimates using the same data. Our results demonstrate that it is not only feasible but also markedly effective, as evidenced by the performance of a Dynamic Linear Model—the Two-Stage Model. When adapted to fit the Italian context, the TSM consistently outperformed traditional polling estimates for nearly all parties in competition across multiple elections. This includes scenarios where polls represent their “best possible attempt,” using aggregated and temporal adjustments. The consistent exception was *Movimento Cinque Stelle*, suggesting that the error stems more from the complexities of this particular party’s support base than from the model itself.

This study contributes to the broader field of electoral forecasting by providing empirical insights into the challenges and potential solutions for improving polling accuracy. It bridges the gap between new democracies—where methods like the TSM have been previously tested—and consolidated democracies with institutional stability but significant electoral volatility. Italy’s case serves as a key example of the “gray space” in democratic systems, characterized by stable institutions yet unpredictable voter dynamics. In this context, our findings emphasize the importance of adapting forecasting methods to account for both structural and methodological complexities.

This study opens several avenues for future research. First, the observed decline in online polling accuracy at larger sample sizes raises critical questions about representativeness and sampling biases that warrant deeper investigation. This is especially relevant given the still recent rise of online pre-electoral polls. Second, the challenges of forecasting in fragmented party systems highlight the necessity for methodological innovations tailored to the complexities of multiparty political environments. It would be especially useful to understand if party system fragmentation affects general political stability by means of producing off-equilibrium expectations. Third, refining models like the TSM and expanding their application across diverse political systems could enhance the robustness of electoral forecasting. The natural step forwards would be to test its application in generally more stable western democracies. Finally, examining the relative performance of alternative methods, such as those based on fundamentals—like economic performance and incumbency—within Italy's shifting political landscape could provide valuable insight for the greater understanding of election forecasting accuracy.

Appendix: Electoral Forecasting in Volatile Party System Settings: Assessing and Improving Pre-Election Poll Predictions in Italy

Table A1. Replication of Table 4 (Model 4 and Model 5), With Interaction Term (Sample Size and Method).

Model	(1) 2008	(2) 2013	(3) 2018	(4) Combined	(5) M4 + Parties
Sample size (squared)	−0.000 (0.000)	−0.000 (0.000)	0.000*** (0.000)	0.000 (0.000)	0.000 (0.000)
Temporal distance (Log)	0.575*** (0.081)	0.934*** (0.050)	0.120*** (0.028)	0.602*** (0.044)	0.630*** (0.037)
Method: Online		0.873 (0.683)	0.179 (0.313)	1.763*** (0.260)	0.472** (0.236)
Method: Phone	0.134 (0.433)	0.099 (0.487)	0.041 (0.155)	0.409** (0.203)	−0.066 (0.165)
Number of parties					0.585*** (0.045)
Sample size: Method:Online		0.000 (0.000)	−0.000*** (0.000)	−0.000* (0.000)	−0.000 (0.000)
Sample size: Method:Phone	0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Constant	1.085** (0.425)	1.711*** (0.479)	2.652*** (0.127)	1.505*** (0.212)	−2.791*** (0.367)
Observations	51	169	147	367	367
Log likelihood	−56.159	−212.165	−89.315	−487.361	−422.155
Akaike Inf. Crit	126.318	442.33	196.63	992.723	864.311
Bayesian Inf. Crit	139.84	470.499	223.544	1027.871	903.364

Source: Author. Dependent variable: MAE (%). Note. The Table only considers pollsters that fielded polls 365 days before election-night. Key: $p < .01$: ***; $p < .05$: **; $p < .01$: ***.

Table A2. Replication of Table 4, With Ordinary Least Squares Regression and Robust Standard Errors.

Model	(1) 2008	(2) 2013	(3) 2018	(4) Combined	(5) M4 + Parties
Sample size (squared)	0.000 (0.000)	0.000* (0.000)	-0.000 (0.000)	-0.000** (0.000)	0.000 (0.000)
Temporal distance (Log)	0.538*** (0.077)	0.926*** (0.046)	0.104*** (0.031)	0.642*** (0.045)	0.660*** (0.036)
Method: Online		0.794*** (0.214)	-0.255 (0.217)	1.490*** (0.193)	0.511*** (0.165)
Method: Phone	0.390 (0.241)	0.580*** (0.164)	-0.164*** (0.062)	0.503*** (0.101)	0.046 (0.085)
Number of parties					0.606*** (0.040)
Constant	0.821*** (0.300)	1.297*** (0.195)	2.954*** (0.098)	1.653*** (0.150)	-2.955*** (0.328)
Observations	51	169	147	367	367
R2	0.543	0.75	0.176	0.435	0.653
Adjusted R2	0.514	0.744	0.152	0.428	0.648
Residual Std. Error	0.457 (df = 47)	0.652 (df = 164)	0.334 (df = 142)	0.865 (df = 362)	0.679 (df = 361)
F Statistic	18.611*** (df = 3; 47)	123.321*** (df = 4; 164)	7.558*** (df = 4; 142)	69.539*** (df = 4; 362)	135.798*** (df = 5; 361)

Source: Author. Dependent variable: MAE (%). Note. The Table only considers pollsters that fielded polls 365 days before election-night. Key: $p < .01$: ***; $p < .05$: **; $p < .01$: ***.

Table A3. Replication of Table 4 (Model 4 and Model 5), With Election Fixed Effects.

	(1) Combined	(2) MI + Parties
Sample size (Squared)	0.00000 (0.00000)	0.00000 (0.00000)
Temporal distance (Log)	0.627*** (0.035)	0.627*** (0.035)
Method: Online	0.510*** (0.183)	0.510*** (0.183)
Method: Phone	0.350*** (0.113)	0.350*** (0.113)
Number of parties		1.587*** (0.114)
Election year: 2013		0.794*** (0.057)
Election year: 2018	0.632*** (0.119)	0.632*** (0.119)
Constant	0.716*** (0.169)	−4.840*** (0.508)
Election fixed effects	Yes	Yes
Pollster random effects	Yes	Yes
Observations	367	367
Log likelihood	−379.666	−380.359
Akaike Inf. Crit	777.331	778.718
Bayesian Inf. Crit	812.480	813.866

Source: Author. Dependent variable: MAE (%). *Note.* The Table only considers pollsters that fielded polls 365 days before election-night. Key: $p < .01$: ***; $p < .05$: **; $p < .01$: ***.

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